Numerical Methods for ODEs: Exercise Week 5

Problem 1:

Show that the improved Euler method from Corollary 33 is consistent with order two.

Problem 2:

Implement the improved Euler method with a constant step size in Python. You can use your implementation of the explicit Euler method as a template. Verify for a suitable example that it provides the expected order of convergence.

Problem 3:

Download the file `euler_adaptive.py` from StudIP. It provides an implementation of the improved Euler / explicit Euler method with adaptive step size control for the IVP in Example 30 in the lecture notes. Devise a way to do a fair comparison of the performance of the adaptive improved Euler versus improved Euler with constant step size. Carry out your analysis and summarize the results.